

Answer Key — Microbes in Our Lives

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **yeast** — yeast respire and releases CO₂ that makes dough fluffy.
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- Q2** (c) **Lactobacillus** — Lactobacillus ferments milk, producing lactic acid (sour curd).
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- Q3** (b) **trapping nitrogen from air to enrich soil** — they fix atmospheric nitrogen, improving soil fertility.
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- Q4** (b) **high salt/sugar stops microbes growing** — high salt/sugar acts as a preservative; microbes can't grow.
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- Q5** (a) **Both true, R explains A** — decomposition recycling waste is exactly how microbes clean the environment.
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- Q6** **biogas** — biogas is mainly methane and carbon dioxide.
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- Q7** **decomposition** — the process is decomposition; the product is manure.
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Short answer

- Q8** Yeast makes dough rise by producing carbon dioxide (and a little alcohol); Lactobacillus turns milk into curd by producing lactic acid.
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- Q9** Soil bacteria/fungi decompose dead matter into nutrient-rich manure; Rhizobium in legume root nodules fixes nitrogen from air, enriching the soil.
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- Q10** Salting/adding sugar (high concentration stops microbes growing) and refrigeration (cold slows microbes); also drying removes the moisture microbes need.
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Long / application

- Q11** Yeast in B respire the sugar and releases carbon dioxide, which inflates the balloon (A has no yeast, so no gas). To prove it is CO₂, pass the gas through lime water — lime water turns milky.
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- Q12** **Friends:** make curd/bread (Lactobacillus, yeast), enrich soil/clean waste (Rhizobium, decomposers), give oxygen (algae) — any two. **Foes:** spoil food (mould on fruit) and cause diseases in plants, animals and humans — any two.
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